

Network Computing

(058172)



POLITECNICO
MILANO 1863

Introduction to Network Computing

Who am I?

Gianni Antichi

- email: gianni.antichi@polimi.it
 - website: <https://gianniantichi.github.io/>
 - office: DEIB, building 20, 3° floor – Campus Leonardo
 - passionate about systems and networking
-
- If you want to meet me, send me an email and we can schedule a time to talk

This course in one slide



Do you know those?

This course in ~~one~~ two slides



Do you know those?

...or those?



This course in ~~one~~ ~~two~~ **three** slides



Do you know those?

...or those?



Have you heard of:

- Cloud
- Online services
- Big Data applications
- ML/AI

This course in ~~one~~ ~~two~~ ~~three~~ **four** slides



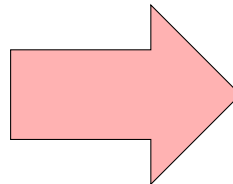
Do you know those?

...or those?



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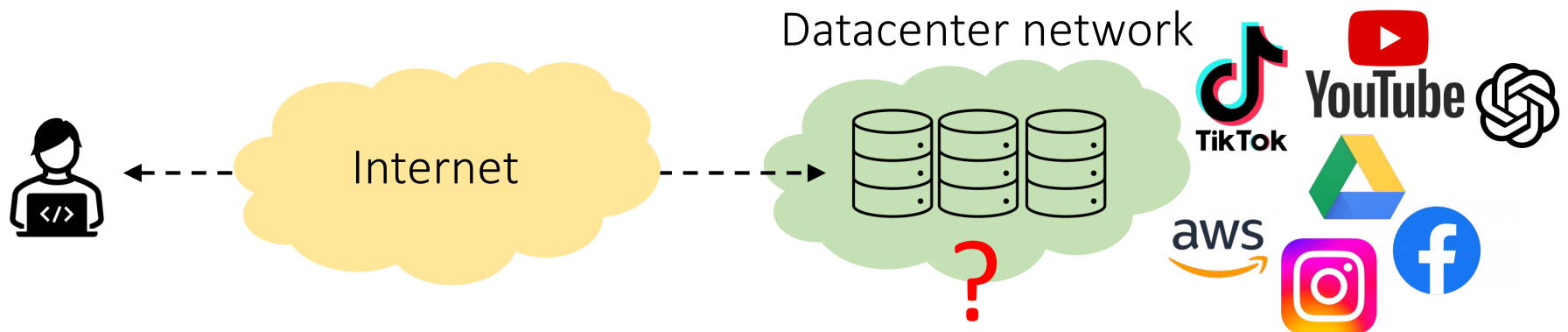


They all run in a datacenter, a facility housing many networked computers that work together to process, store, and share data.

This course in ~~one~~ ~~two~~ ~~three~~ ~~four~~ five slides

The goal for this course is to reach an understanding of networking concepts and protocols within datacenter networks

WHY? because most major companies rely heavily upon datacenters as a central component in delivering online services

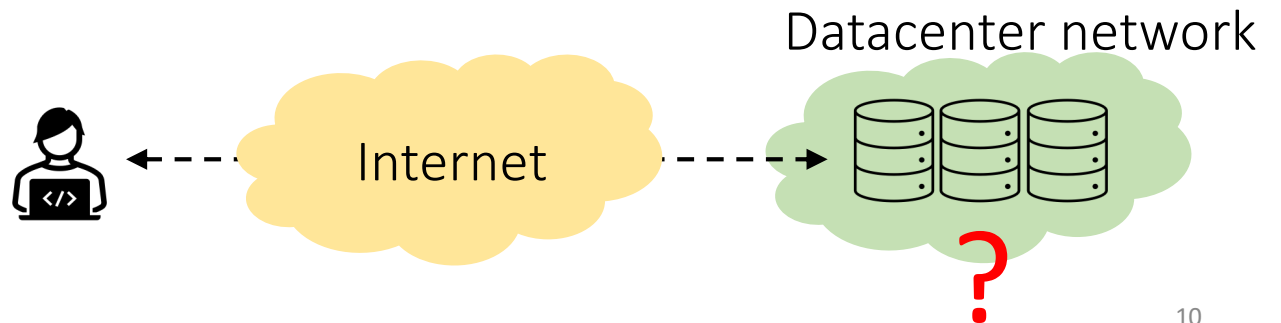


Prerequisites

- This course assumes a basic understanding of topics in networking, such as packet-switching, routing and socket programming
- It is also helpful to know how to program
- For a refresher, you could look at the textbook “Computer Networking: A Top-Down Approach” by J. Kurose and K. Ross or “Computer Networks: A Systems Approach” by L. Peterson and B. Davie

Interaction during lectures

- Share your questions/comments/wishes publicly
 - I love interaction!
 - We have a forum as well!
- No such a thing as a *silly question*
- It is *your responsibility* to ask for clarifications!



Basic course information

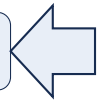
- All up-to-date informations can be found on the course webpage available on WeBeep (search for 058172 - NETWORK COMPUTING)
- Course material
 - Unfortunately, there is no single book that cover all the stuff you are going to see in this course
 - ...but you can find lots of useful stuff here: <https://sdn.systemsapproach.org/>
 - + Slides
 - + Papers I will share with you on the course website

Teaching approach

- Lectures
- Paper reading
- Lab

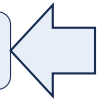
Teaching approach

- Lectures
- Paper reading
- Lab
- ...to bring new concepts on the table



Teaching approach

- Lectures
 - Paper reading
 - Lab
-
- ...to bring new concepts on the table
 - ...to build skills for critically engaging with academic papers



Teaching approach

- Lectures
- Paper reading
- Lab



- ...to bring new concepts on the table
- ...to build skills for critically engaging with academic papers
- ...to gain practical experience

Teaching approach

- Lectures
- Paper reading
- Lab



Farbod Shahinfar
<https://fshahinfar1.github.io/>
farbod.shahinfar@polimi.it

- ...to bring new concepts on the table
- ...to build skills for critically engaging with academic papers
- ...to gain practical experience

Teaching approach

- Lectures

- Paper reading

- Lab

- ...to bring new concepts on the table
- ...to build skills for critically engaging with academic papers
- ...to gain practical experience

Golden rule: use the lectures and lab time to learn! Ask as much as you want

Exam

- Oral 😊
- Questions about what was covered in the lectures (as simple as that 😊)
- I will evaluate you based on your understanding of the topics
 - It is fine not to remember every single bit of a packet header
 - It is less ok if you do not remember why we need a specific datacenter functionality that we discussed in a two hours lecture!

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This will bring you as far as to 21/30L
I am afraid 😊

Exam + mini project

- To get additional points you need to get your hands dirty 😊

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 - We emphasize replication of a prior result to introduce the complexity and rigor that are central to an engineering career where understanding and extending existing code is far more common than developing systems from scratch

Exam + mini project

We say try because you might not be able to! Whether you can or can't, both are useful scientific results

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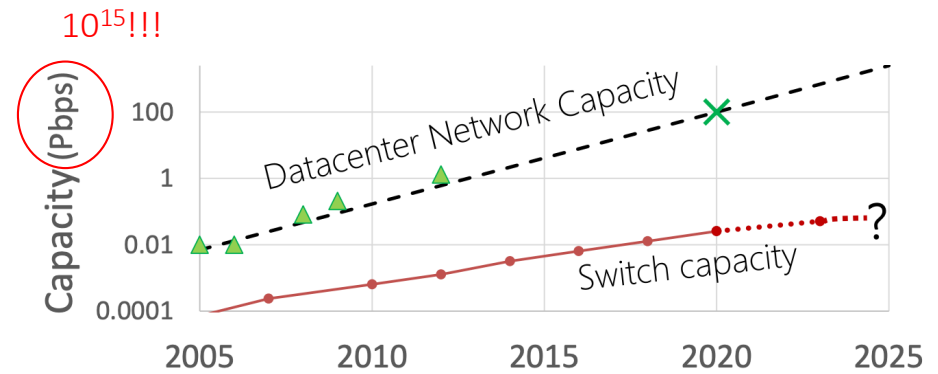
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- A group-based project on **trying** to replicate an experimental result from a networking research paper
 - We emphasize replication of a prior result to introduce the complexity and rigor that are central to an engineering career where understanding and extending existing code is far more common than developing systems from scratch
- We will provide a set of papers from which you can choose
 - You will write a report that has to be submitted 7 days before the date of your exam
 - More info will be shared by Farbod during the lab sessions

Module content

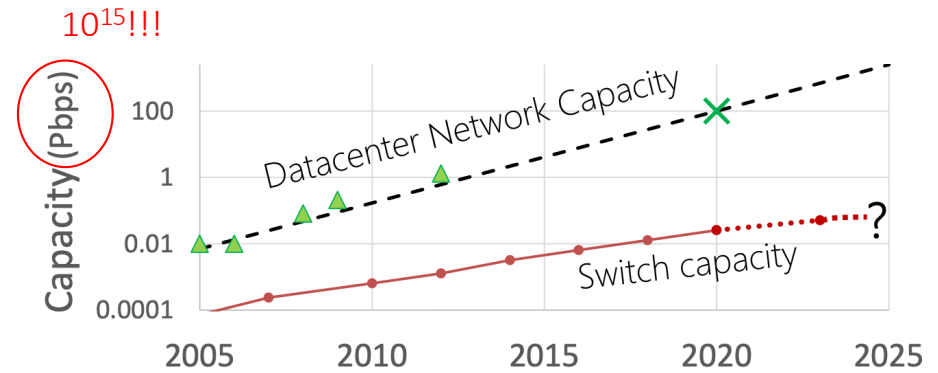
- Datacenter networks and their challenges



Sirius: A Flat Datacenter Network with Nanosecond Optical Switching (ACM SIGCOMM)

Module content

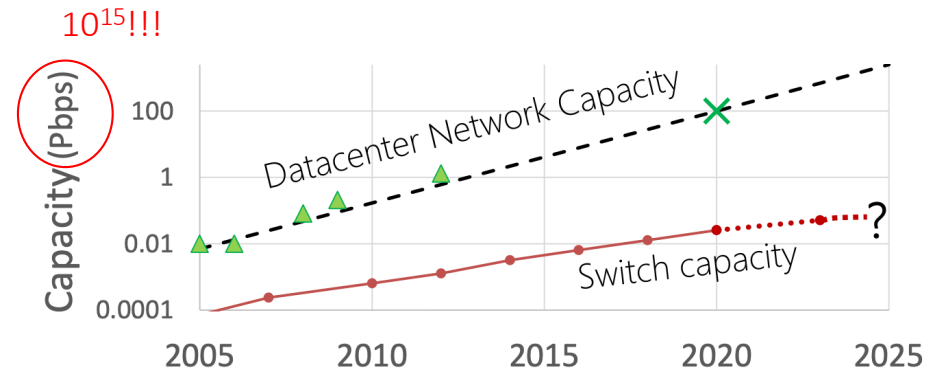
- Datacenter networks and their challenges
- 1 petabit is 1000 terabits
- 1 terabit is 1 trillion bits



Sirius: A Flat Datacenter Network with Nanosecond Optical Switching (ACM SIGCOMM)

Module content

- Datacenter networks and their challenges
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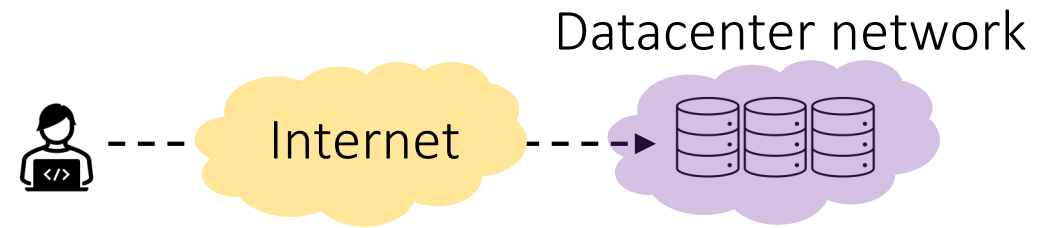


Sirius: A Flat Datacenter Network with Nanosecond Optical Switching (ACM SIGCOMM)

An average 4K video is approximately 120Gb/hour. This mean 1 petabit of storage could hold around 8500 hours: you need almost one-year non-stop to see everything!

Module content

- Datacenter networks and their challenges



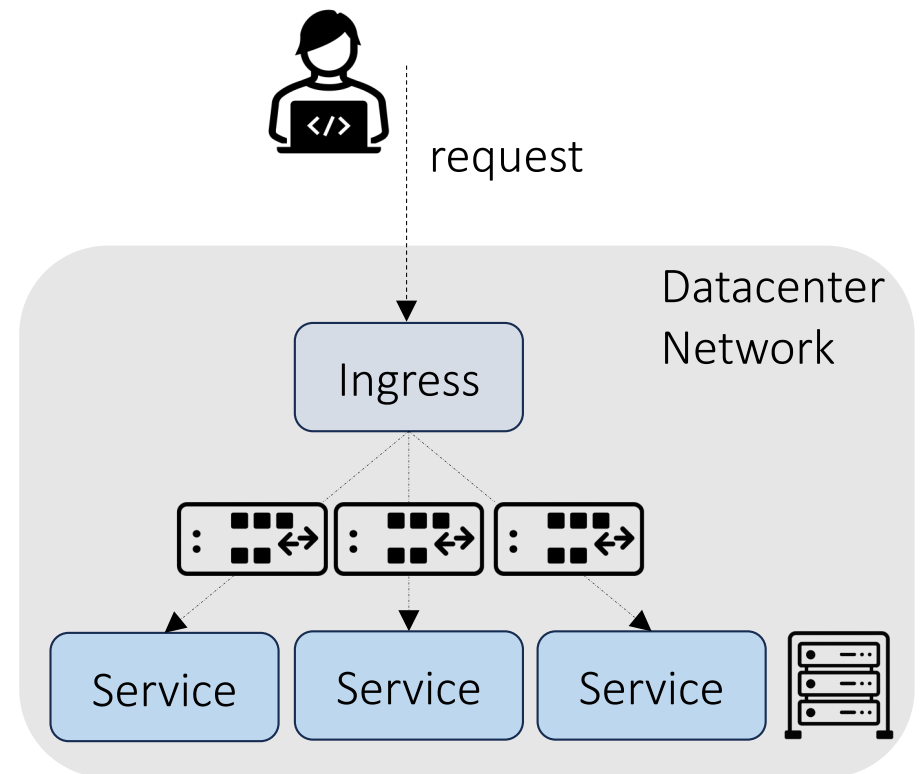
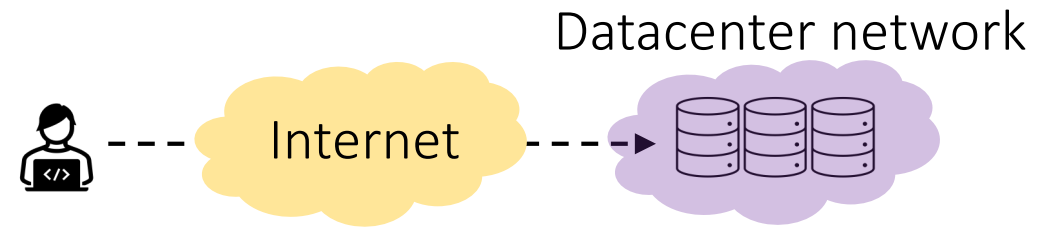
Module content

- Datacenter networks and their challenges
 - Techniques to handle terabit-scale traffic

L3 load balancing

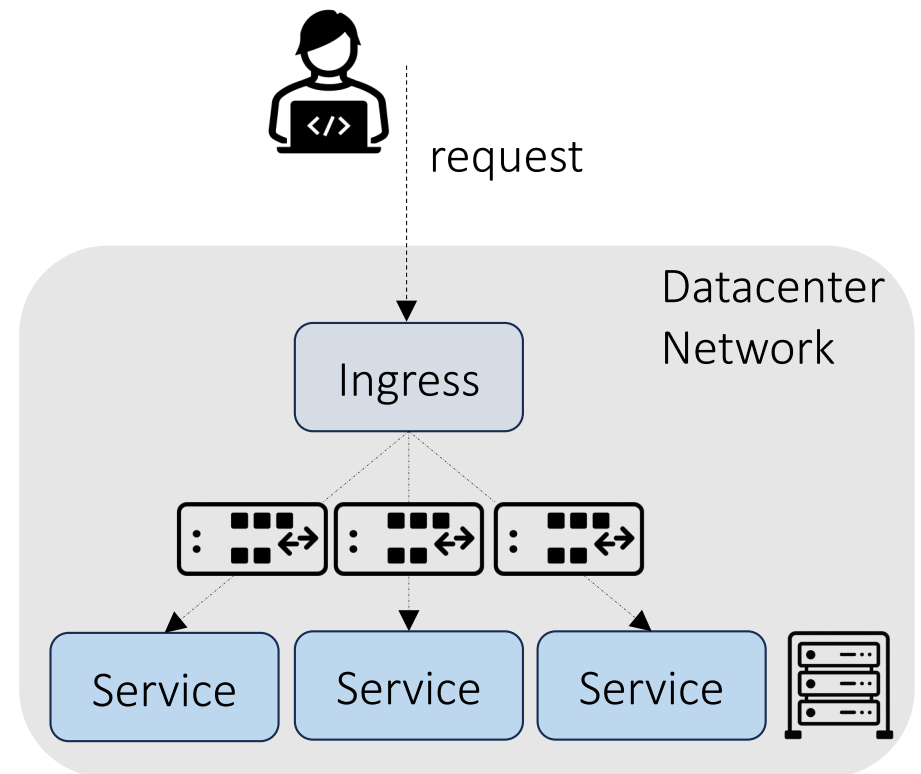
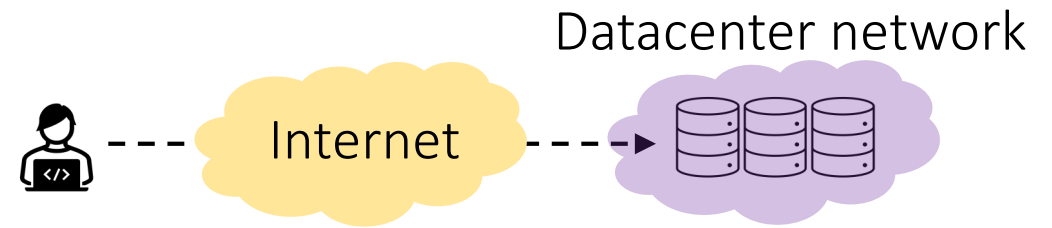
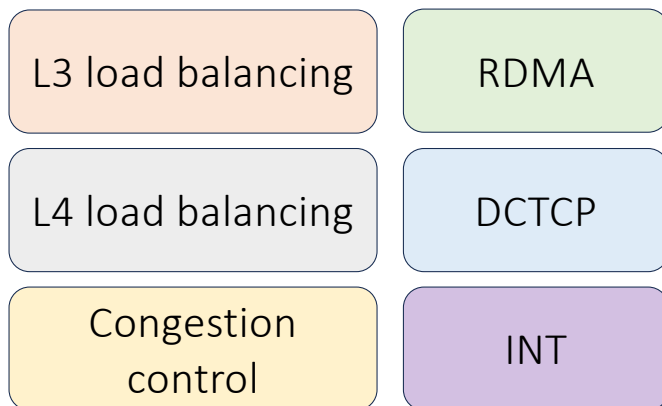
L4 load balancing

Congestion
control



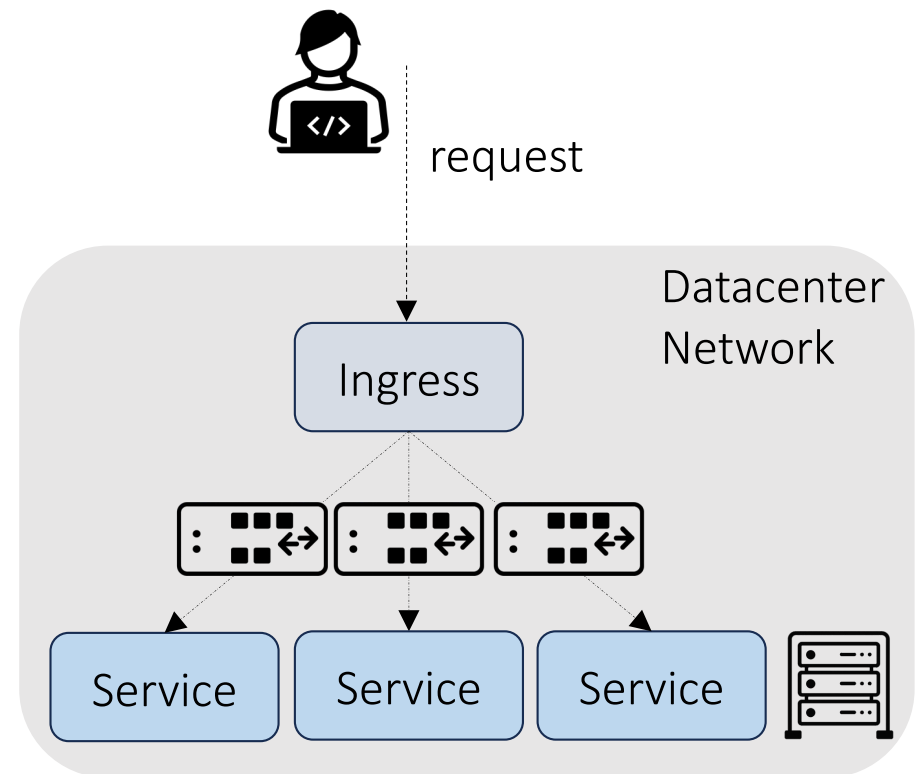
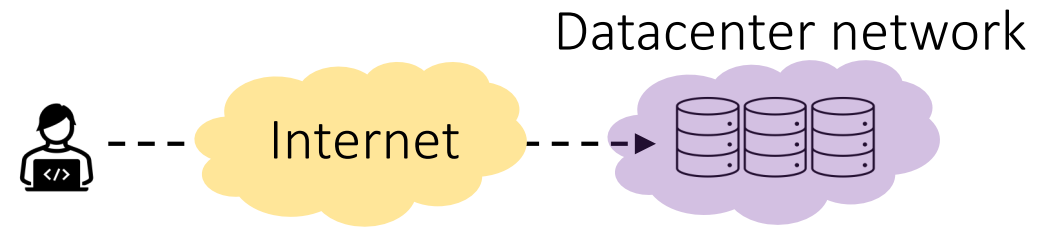
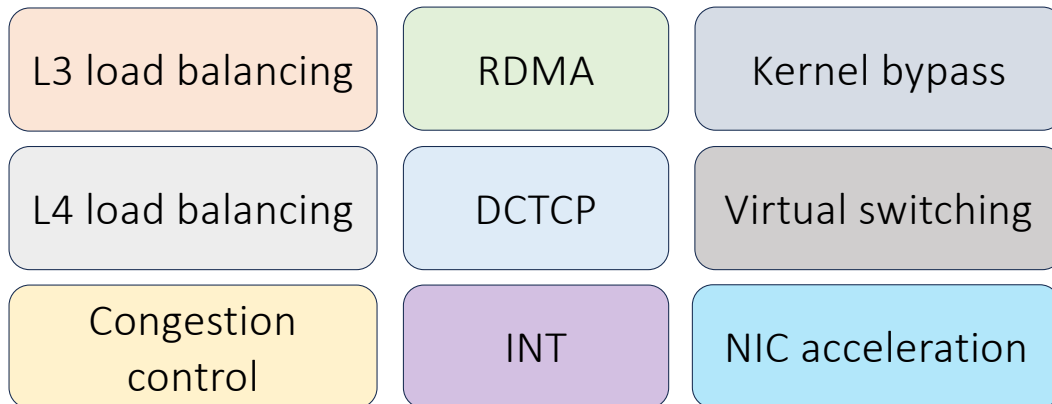
Module content

- Datacenter networks and their challenges
 - Techniques to handle terabit-scale traffic
 - Protocols used



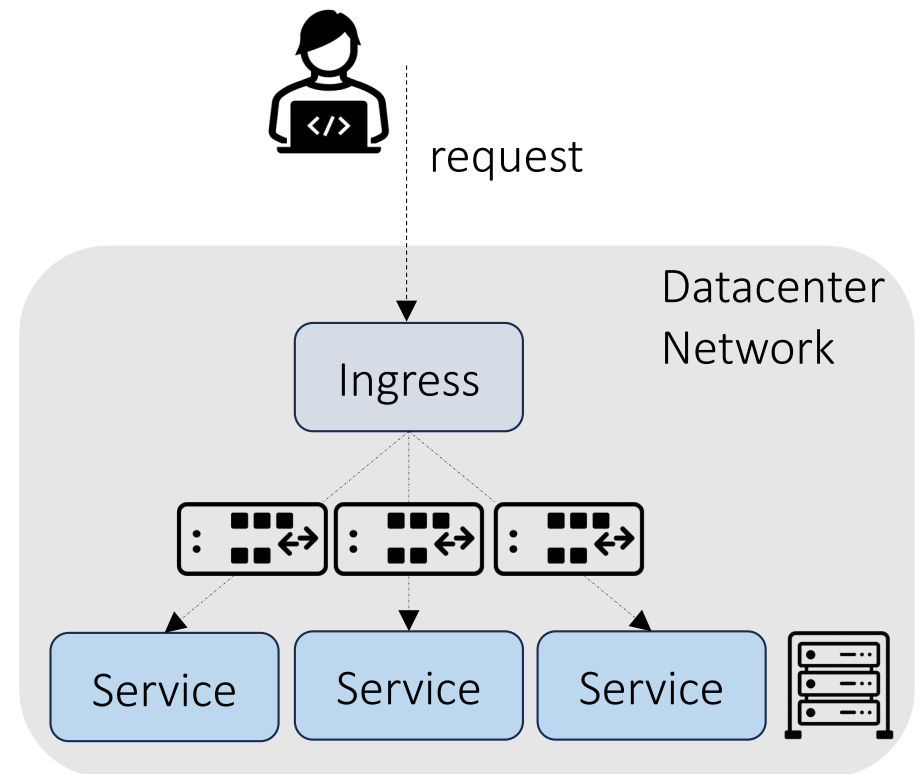
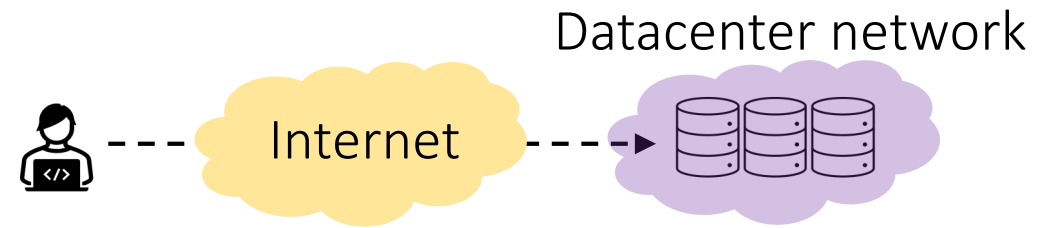
Module content

- Datacenter networks and their challenges
 - Techniques to handle terabit-scale traffic
 - Protocols used
 - End-host network stack specialization



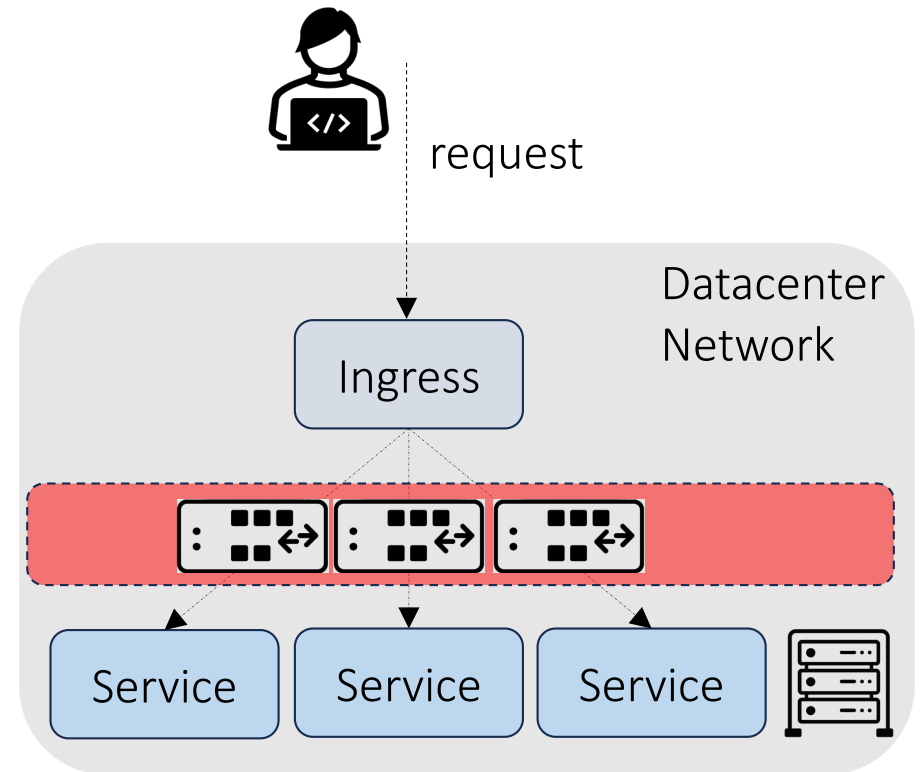
Module content

- This is a highly fertile topic because:
 - Datacenters host most popular applications
 - Datacenters can deviate from standards
 - Modern networking hardware and software can now be programmatically controlled



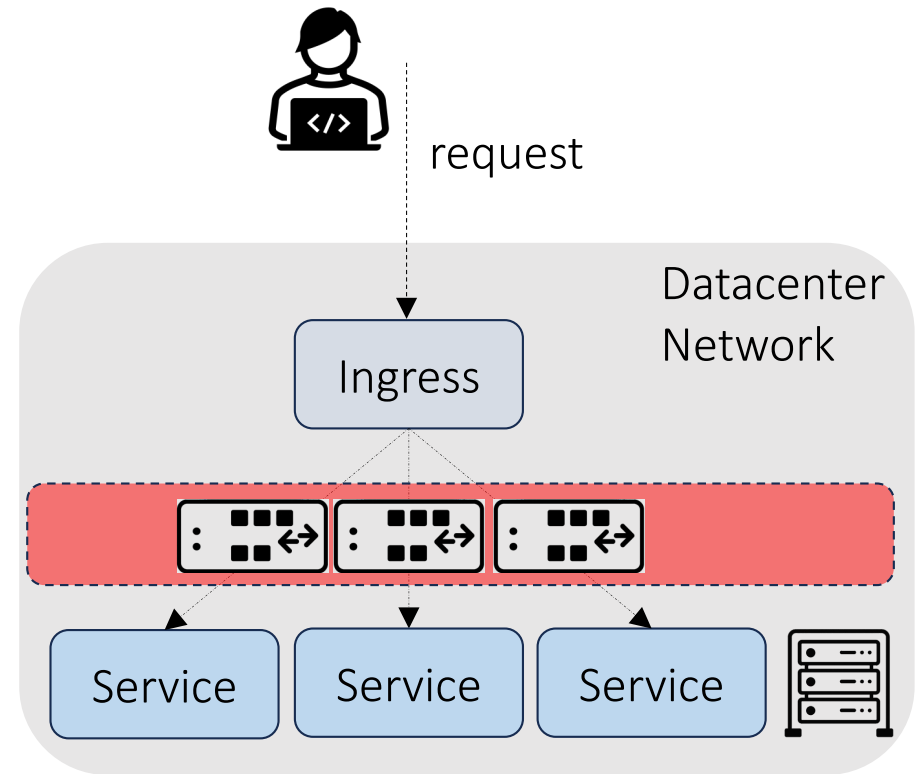
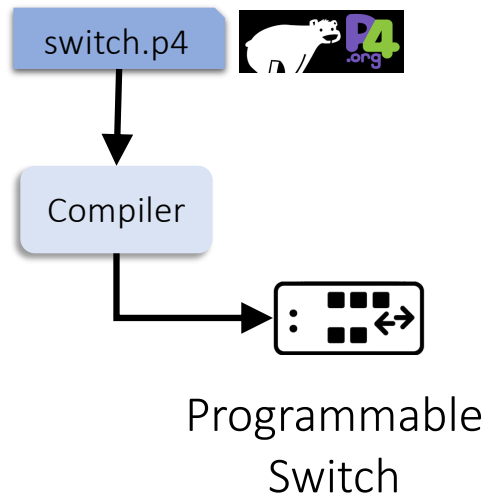
Example 1

- Once upon a time you could *program* switches using their available APIs



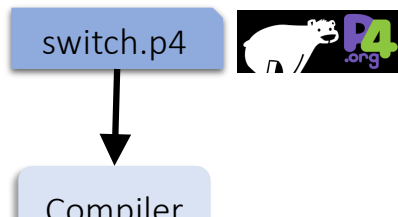
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- Now...



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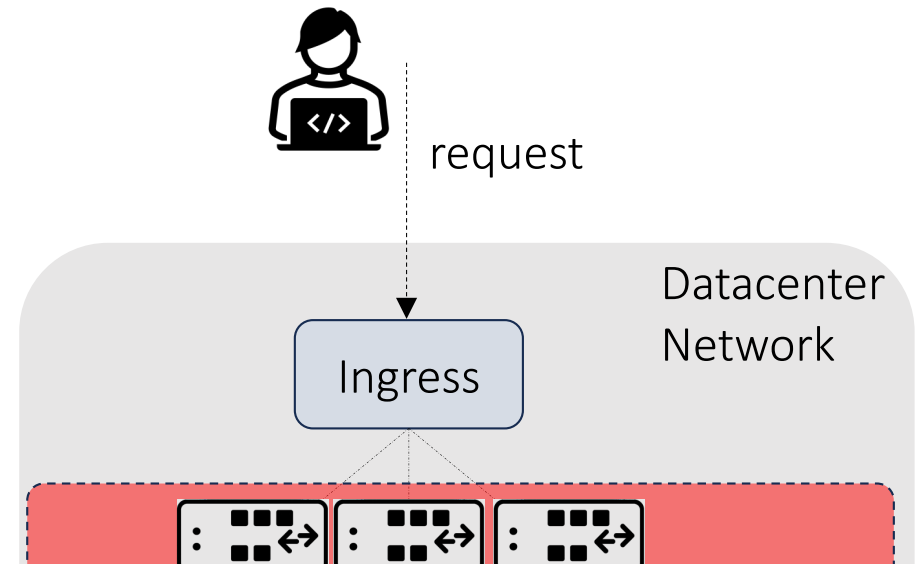


Asterfusion releases 400G P4 Programmable Switch by Intel Tofino 2

ARISTA



Asterfusion data technologies · Follow
3 min read · Nov 8, 2023



Using Trio – Juniper Networks’ Programmable Chipset – for Emerging In-Network Applications

Mingran Yang[†], Alex Baban*, Valery Kugel*, Jeff Libby*, Scott Mackie*,
Swamy Sadashivaiah Renu Kananda*, Chang-Hong Wu*, Manya Ghobadi[†]

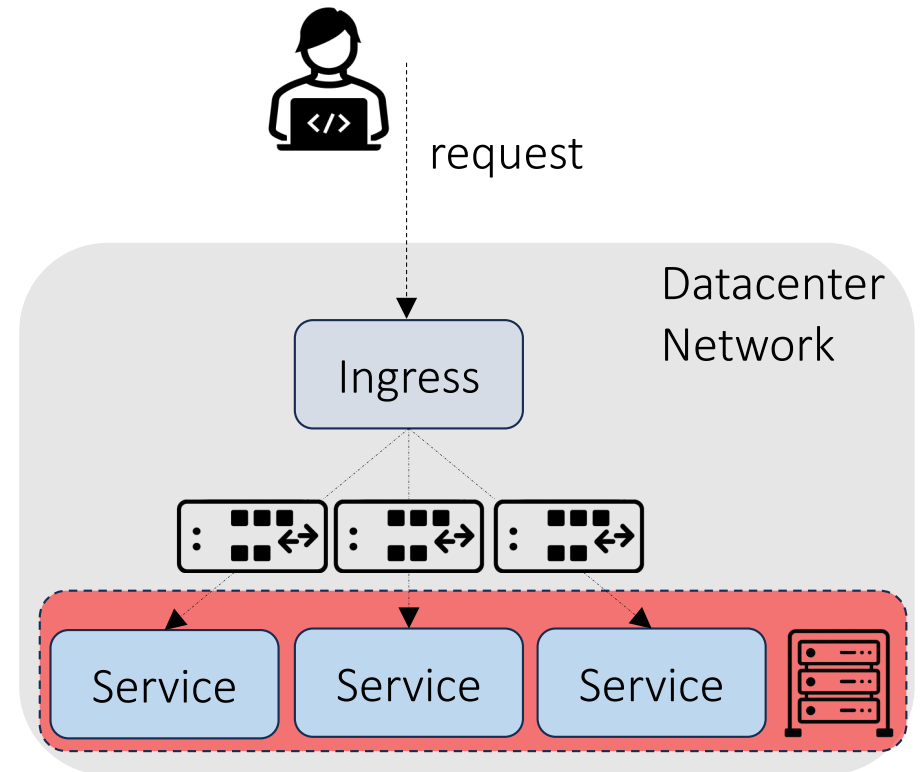
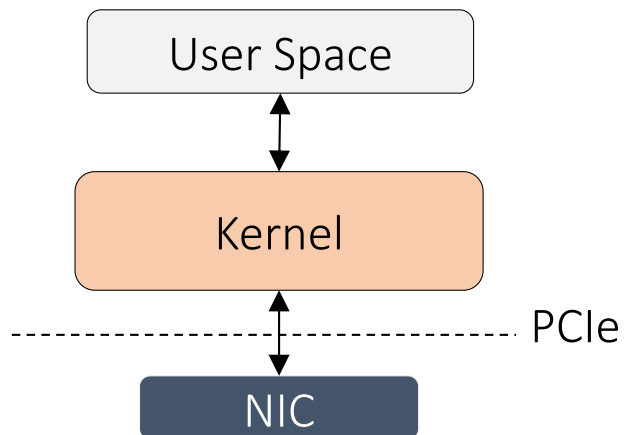
[†]Massachusetts Institute of Technology *Juniper Networks

7170B Series Programmable Data Center Switches

Data Sheet

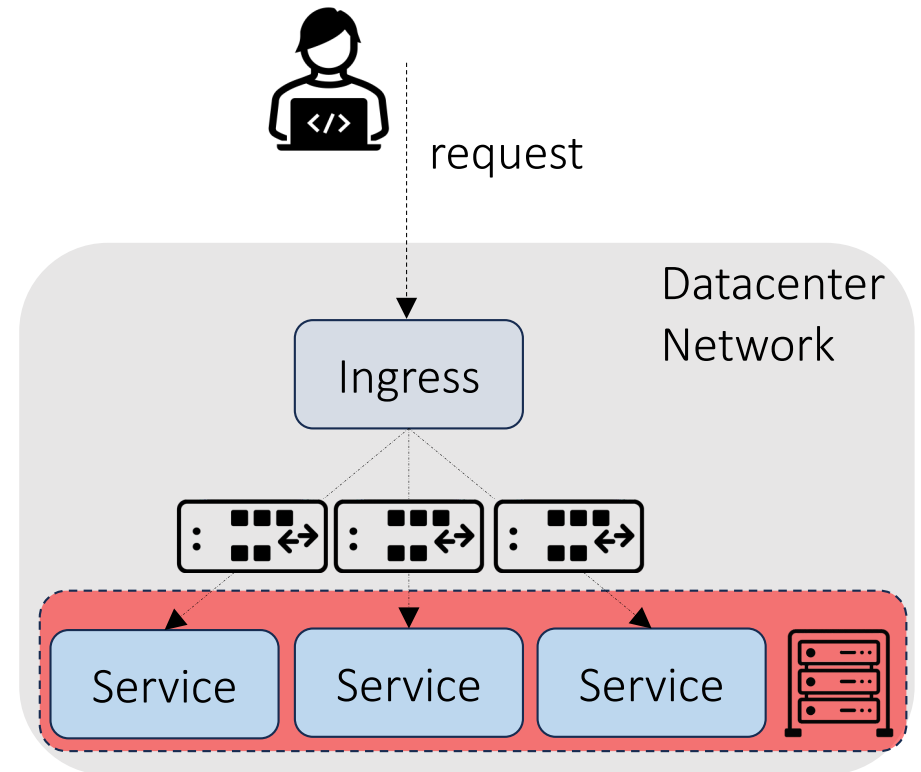
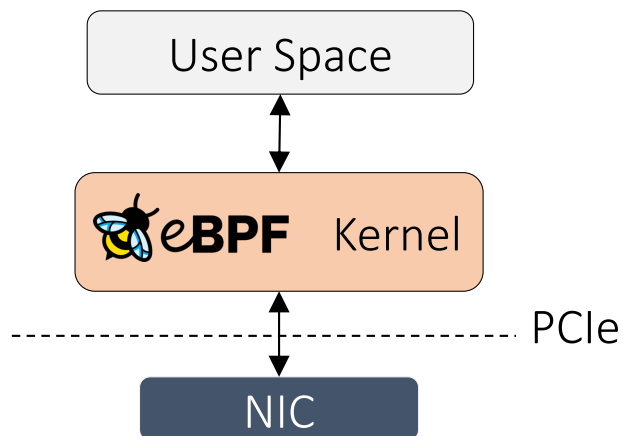
Example 2

- eBPF is a technology that allows specially written programs to safely run in the kernel
 - No changes to kernel source
 - No need to recompile the kernel



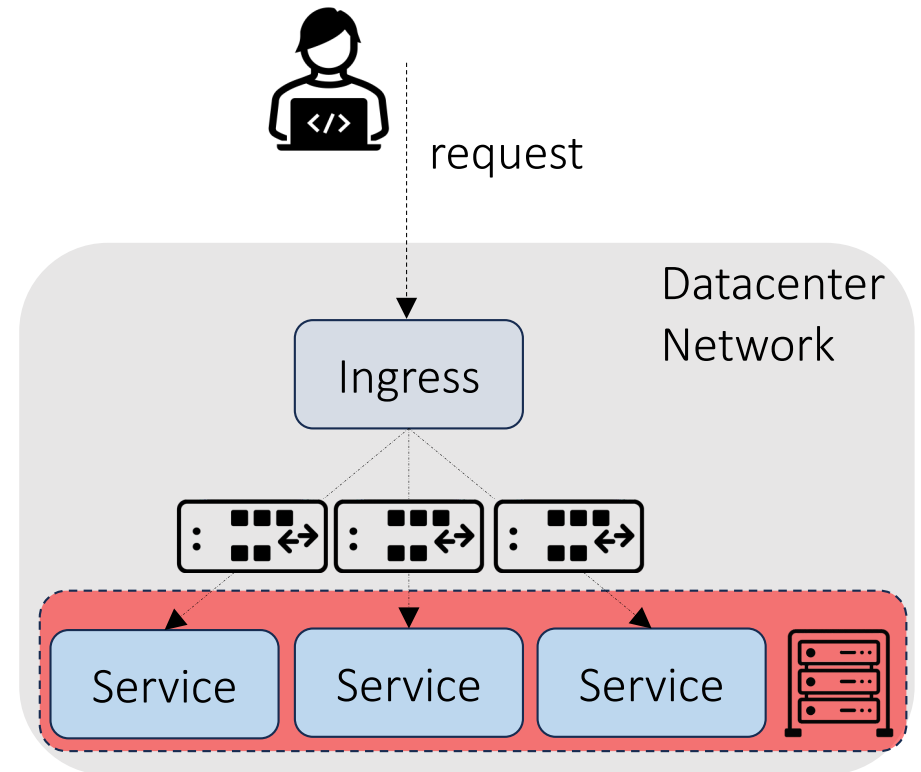
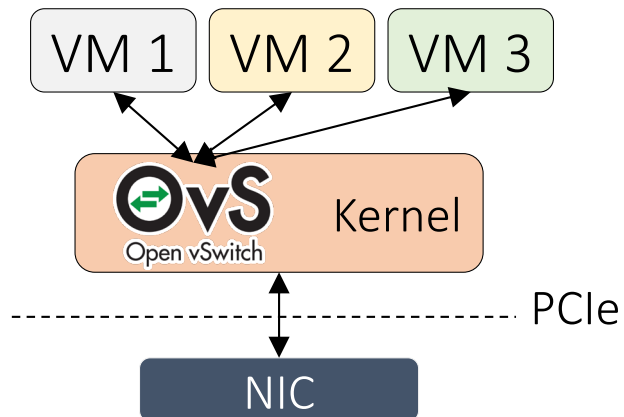
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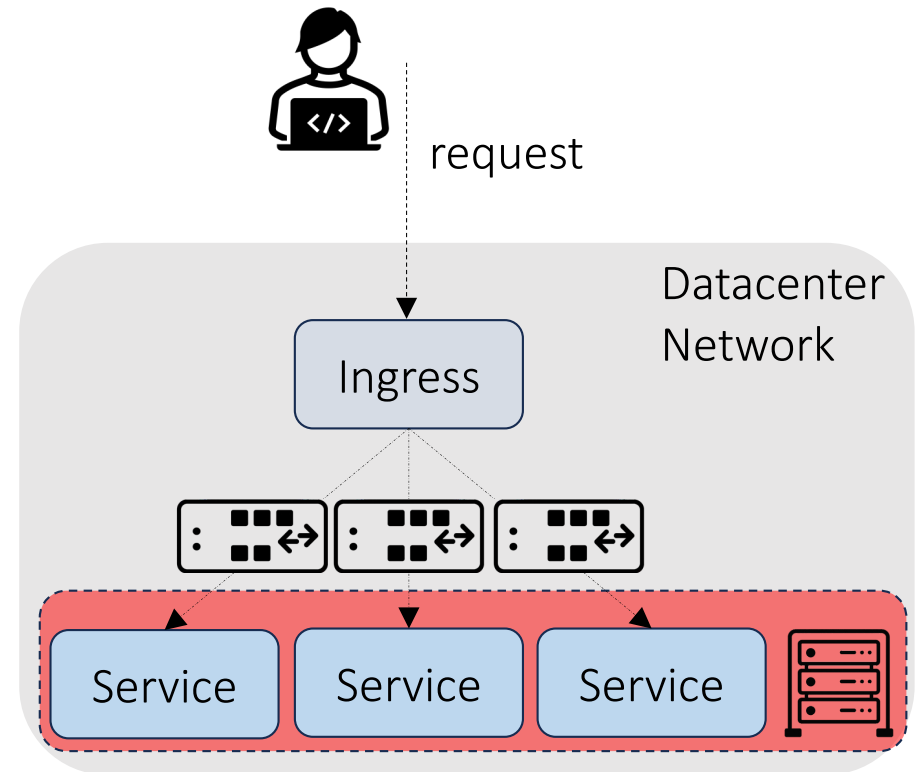
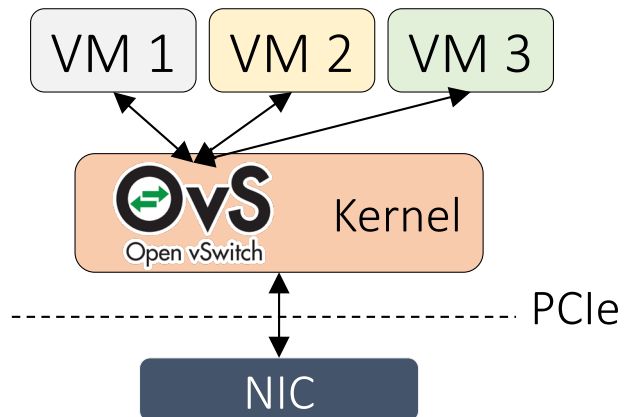
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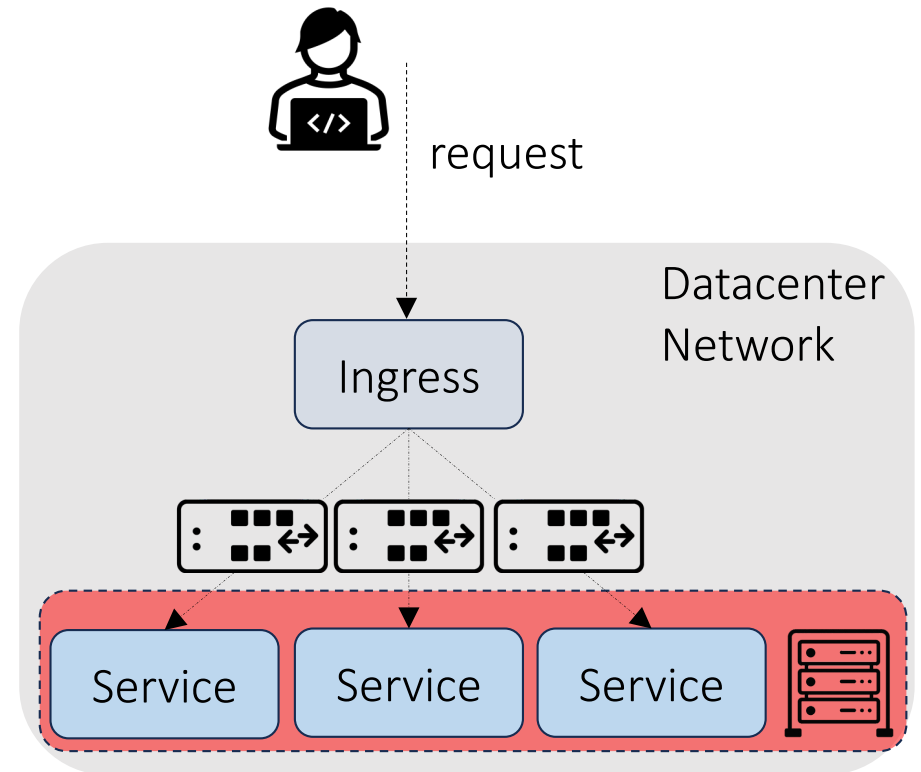
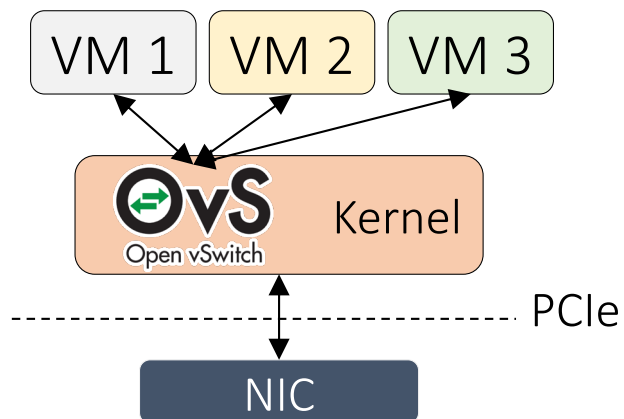
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OVS is part of the NSX product for datacenter virtualization used by thousands of VMware customers!

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OVS is part of the NSX product for datacenter virtualization used by thousands of VMware customers!

"The high bar to acceptance discourages developers from proposing features that would require experimentation or refinement, and Linux developers do reject some new OVS features"

Overall

- In this course, we will try to find the right balance between
 - What the industry actually does in operation
 - What the research community around the world proposes
- We will have invited speakers (a couple or three) from Industry that will be able to share their knowledge and insights with you 😊
 - Microsoft Research and Meta confirmed so far
 - We might get one more from Cloudflare, Google or alike but it is not confirmed yet
 - Grasp this opportunity with two hands and ask many questions!

Should you take this course?



Should you take this course?

- YES 😊 (if you like computer networks)

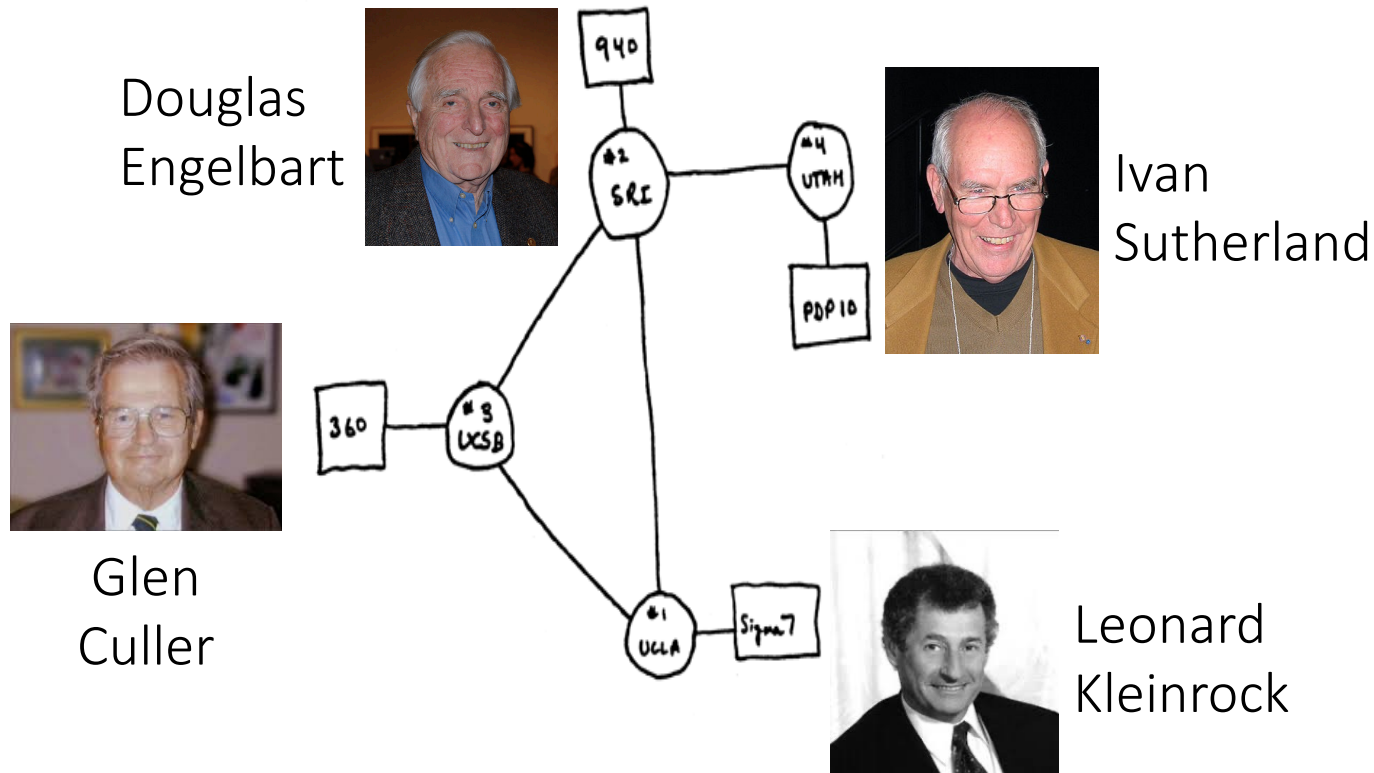


Should you take this course?

- YES 😊 (if you like computer networks)
- Maybe not if:
 - You hate programming
 - You hate reading papers about new advances in network systems
 - You expect 10+ years of exam history 😊



Networking in the early days

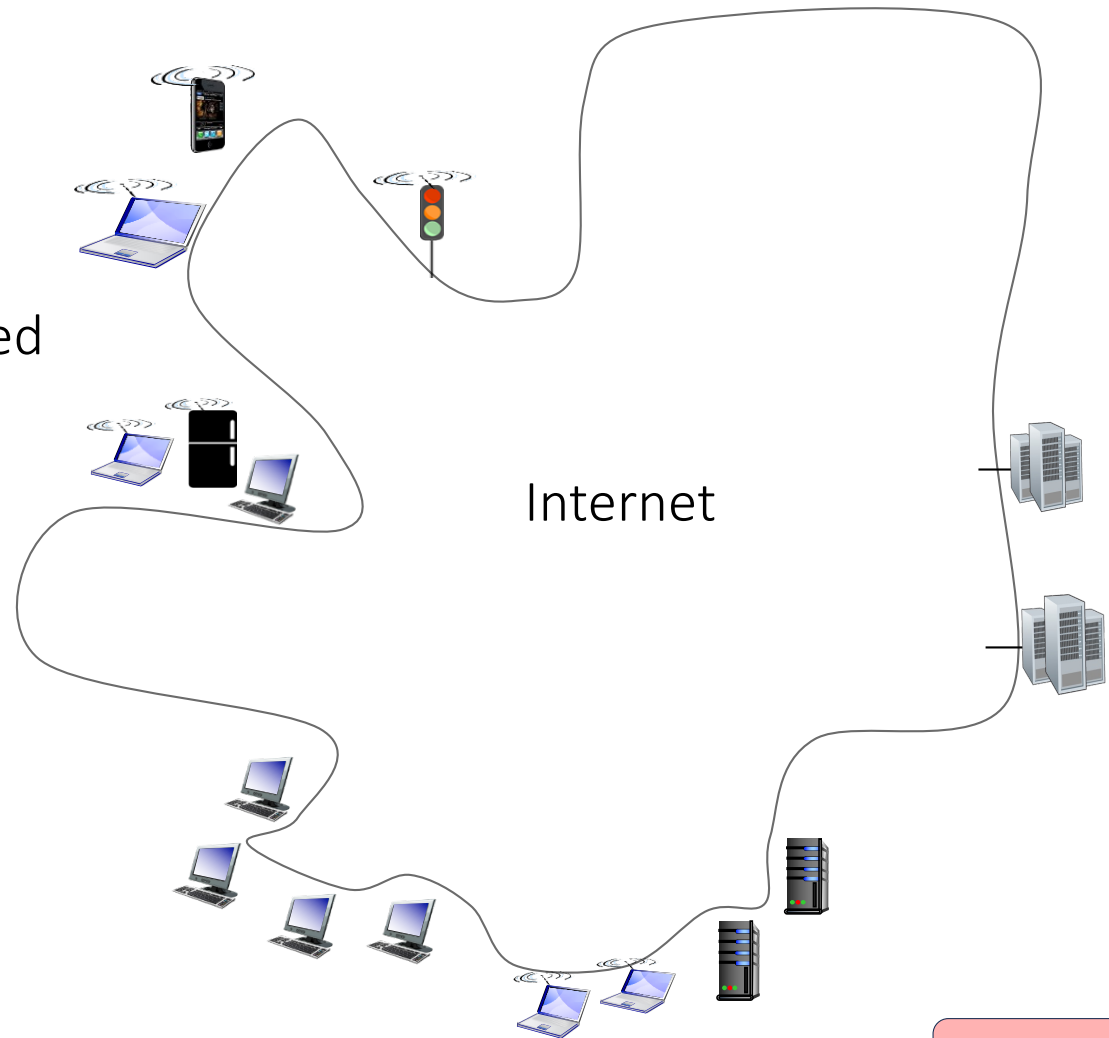


1964

Networking now..



Billions of connected
computing devices



2023

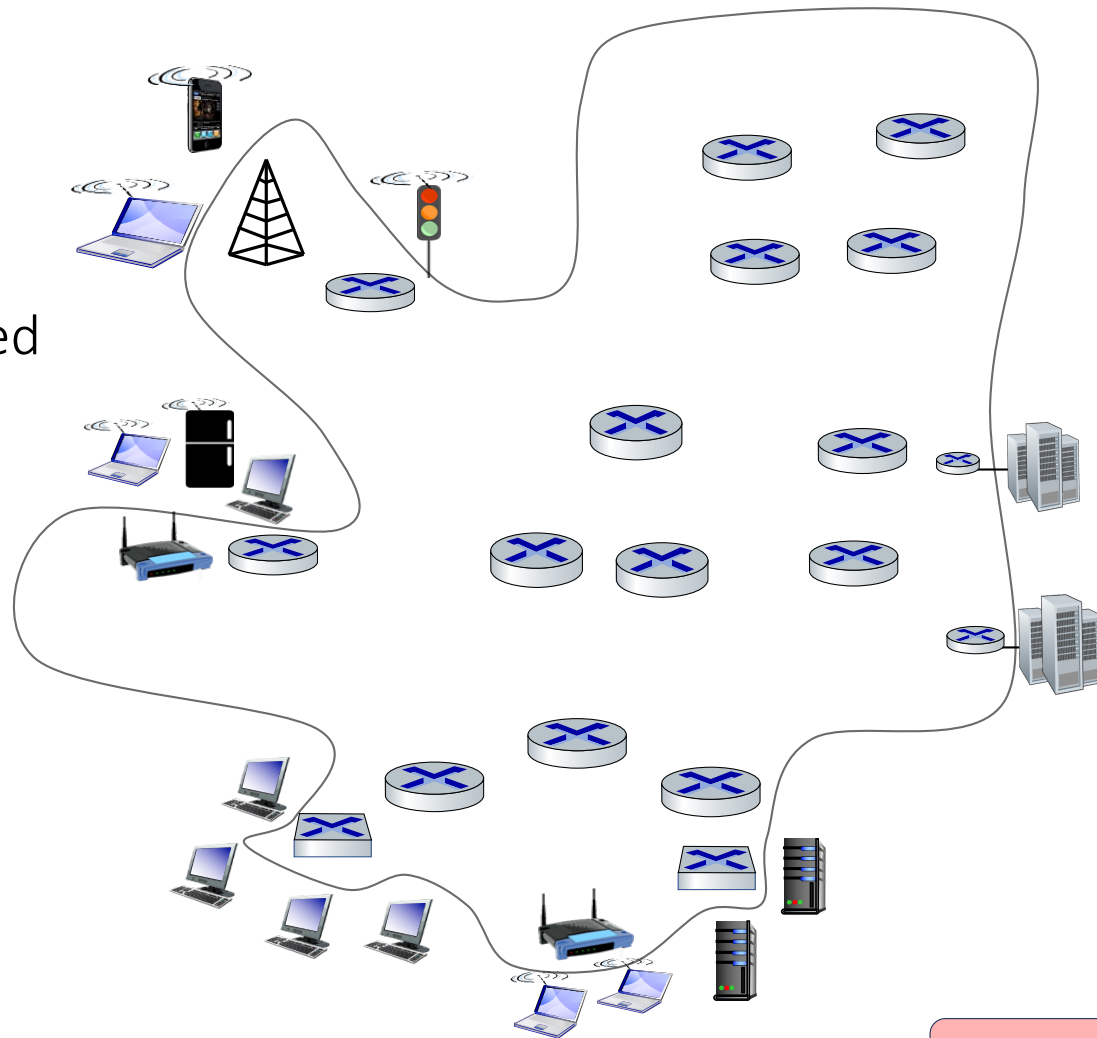
Networking now..



Billions of connected computing devices



Switches/Routers forwarding data



2023

Networking now..



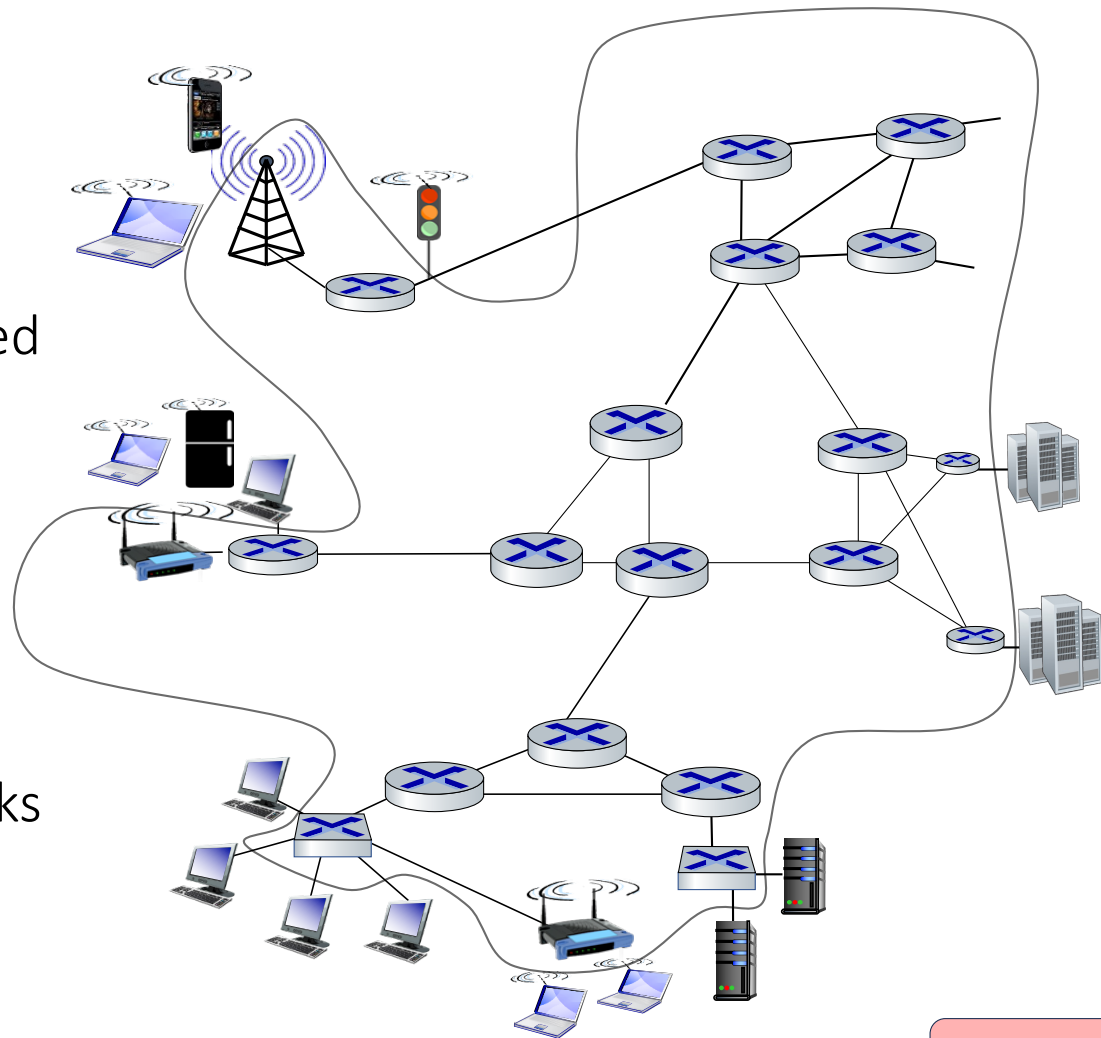
Billions of connected computing devices



Switches/Routers forwarding data



Communication links



2023

Networking now..



Billions of connected computing devices



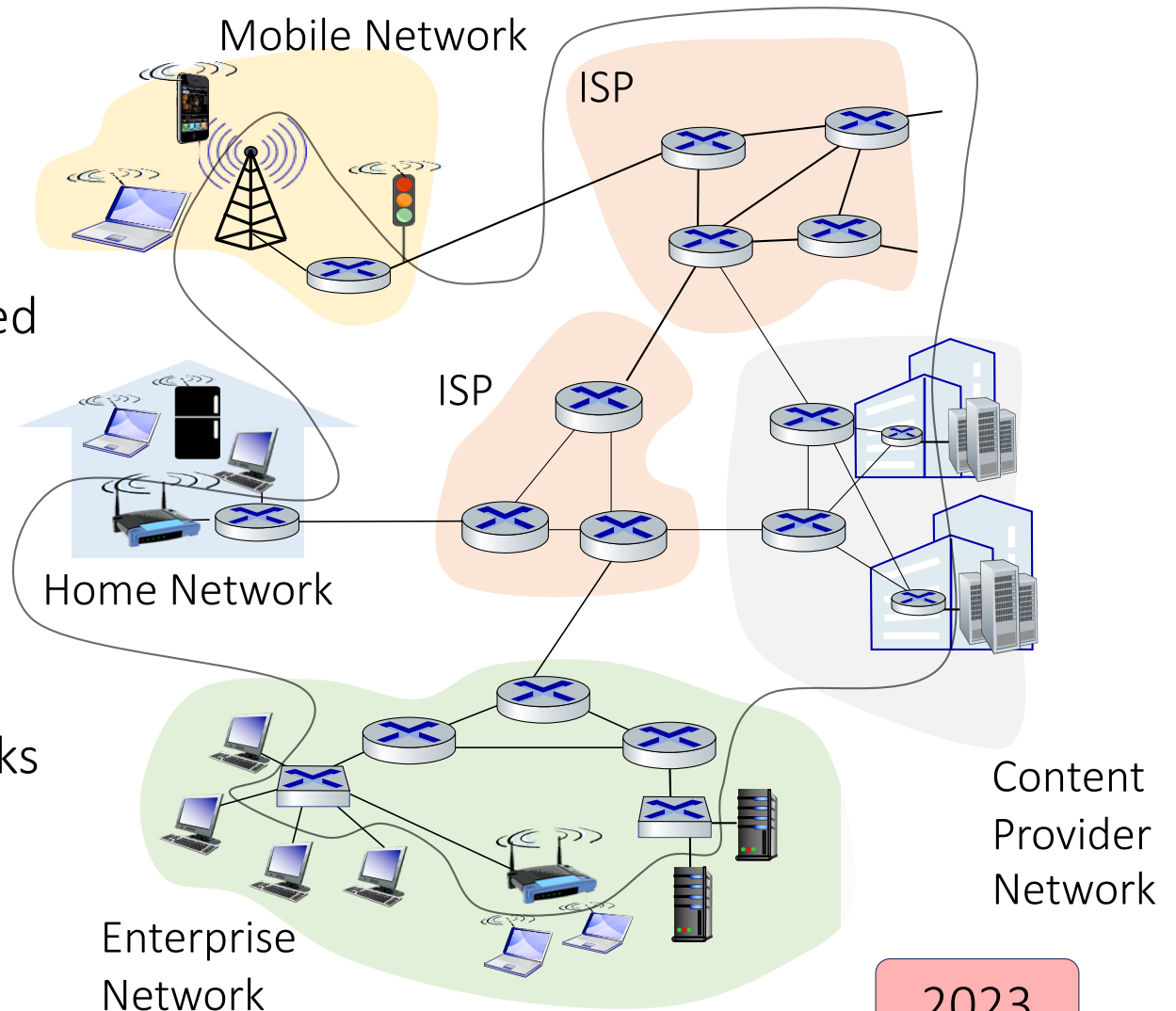
Switches/Routers forwarding data



Communication links



Networks



2023

Networking now..



Billions of connected computing devices



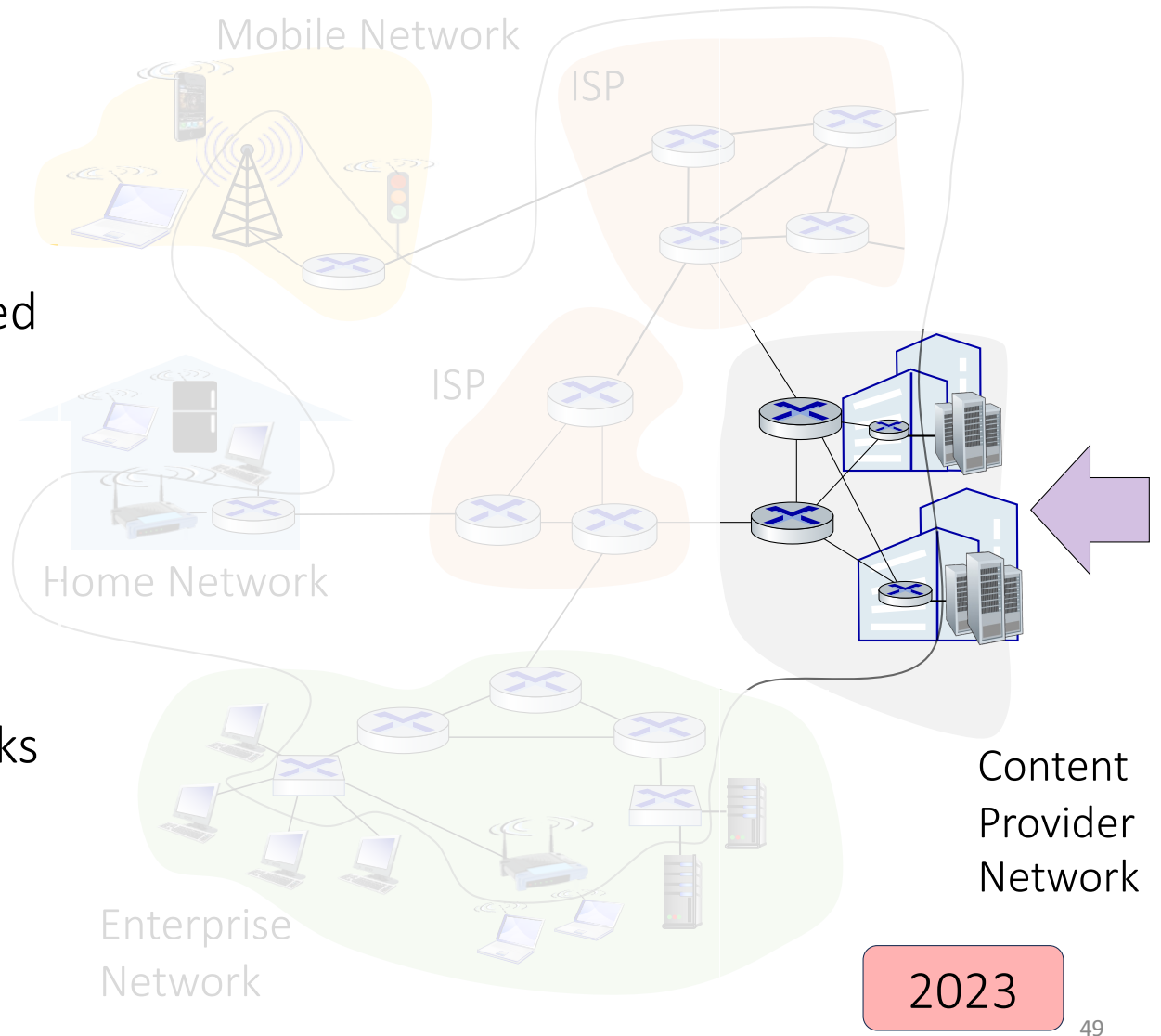
Switches/Routers forwarding data



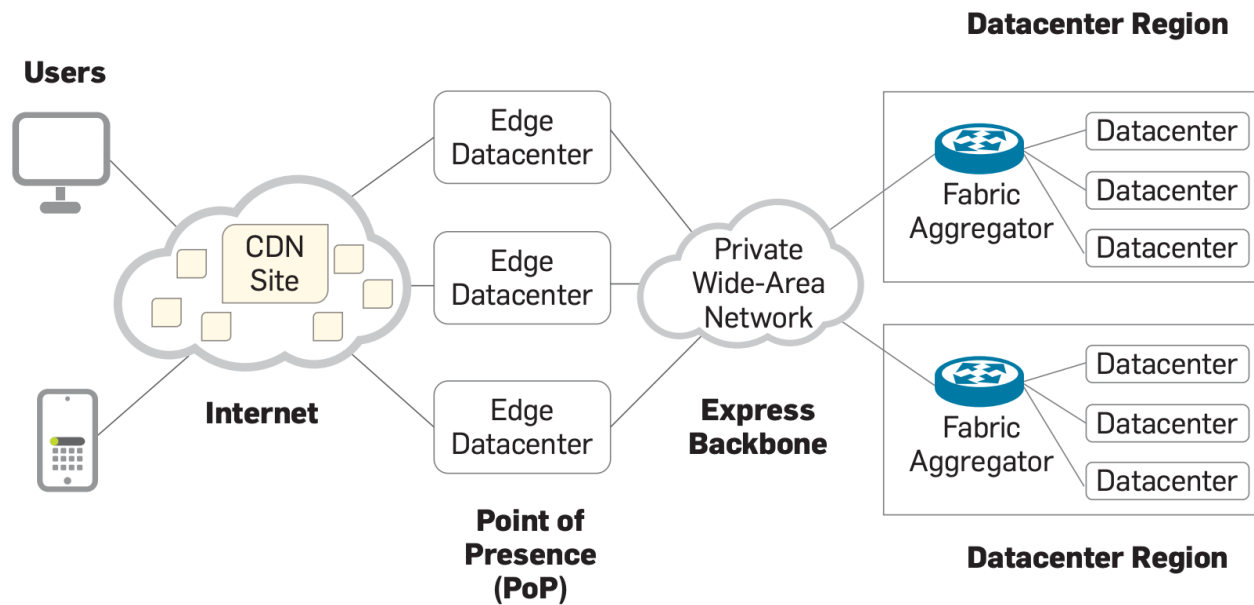
Communication links



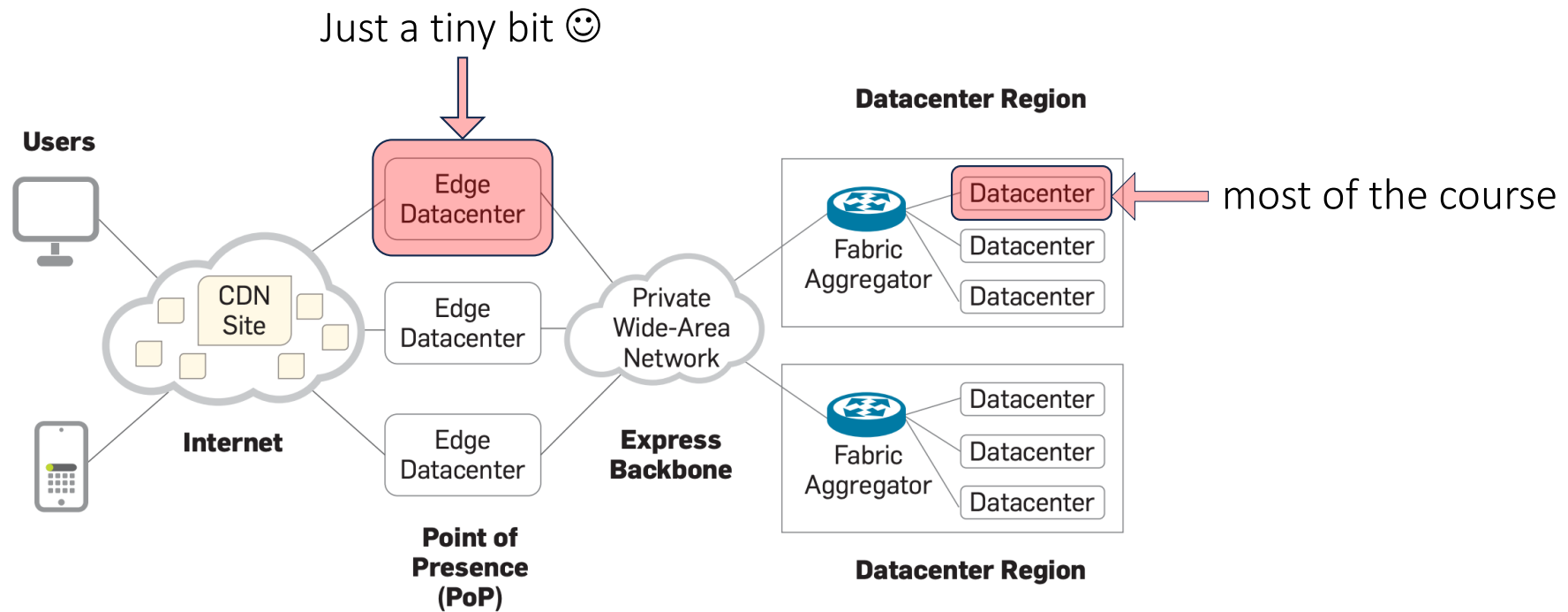
Networks



A real-world example: Meta's Infrastructure



A real-world example: Meta's Infrastructure



What do you think a datacenter is in practice?

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<https://www.linkedin.com/pulse/cloud-really-just-someone-elses-computer-marcelo-oliveira/>

What do you think a datacenter is in practice?



A collection of many networked devices. More in the next lecture 😊

Research papers

- Research papers will be an important source of information during the course
- They basically form the *Network Computing* course book
- Warning: research papers published at academic conferences can also provide information about systems in production!

Scaling Memcache at Facebook

Rajesh Nishtala, Hans Fugal, Steven Grimm, Marc Kwiatkowski, Herman Lee, Harry C. Li,
Ryan McElroy, Mike Paleczny, Daniel Peek, Paul Saab, David Stafford, Tony Tung,
Venkateshwaran Venkataramani

{rajeshn,hans}@fb.com, {sgrimm, marc}@facebook.com, {herman, hcli, rm, mpal, dpeek, ps, dstaff, ttung, veeve}@fb.com

Facebook Inc.

Abstract: Memcached is a well known, simple, in-memory caching solution. This paper describes how Facebook leverages memcached as a building block to construct and scale a distributed key-value store that supports the world's largest social network. Our system handles billions of requests per second and holds trillions of items to deliver a rich experience for over a billion users around the world.

however, web pages routinely fetch thousands of key-value pairs from memcached servers.

One of our goals is to present the important themes that emerge at different scales of our deployment. While qualities like performance, efficiency, fault-tolerance, and consistency are important at all scales, our experience indicates that at specific sizes some qualities require more effort to achieve than others. For exam-

Sailfish: Accelerating Cloud-Scale Multi-Tenant Multi-Service Gateways with Programmable Switches

Tian Pan[†], Nianbing Yu[†], Chenhao Jia[†], Jianwen Pi[†], Liang Xu[†], Yisong Qiao[†], Zhiguo Li[†],
Kun Liu[†], Jie Lu[†], Jianyuan Lu^{†□}, Enge Song[†], Jiao Zhang^{*}, Tao Huang^{*}, Shunmin Zhu^{*†□}

[†]Alibaba Group

^{*}Purple Mountain Laboratories

^{*}Tsinghua University

ABSTRACT

The cloud gateway is essential in the public cloud as the central hub of cloud traffic. We show that horizontal scaling of software gateways, once sustainable for years, is no longer future-proof facing the massive scale and rapid growth of today’s cloud. The root cause is the stagnant performance of the CPU core, which is prone to be overloaded by heavy hitters as traffic growth goes far beyond Moore’s law. To address this, we propose *Sailfish*, a cloud-scale multi-tenant multi-service gateway accelerated by programmable switches. The new challenge is that large forwarding tables due to multi-tenancy cannot be fit into the limited on-chip memories. To this end, we devise a multi-pronged approach with (1) hardware/software co-design for table sharing, (2) horizontal table splitting among gateway clusters, (3) pipeline-aware table compression for a single node. Compared with the x86 gateway of a similar price, *Sailfish* reduces latency by 95% ($2\mu s$), improves throughput by more than 20x in bps (3.2Tbps) and 71x in pps (1.8Gpps) with packet length $< 256B$. *Sailfish* has been deployed in Alibaba Cloud for more than two years. It is the first P4-based cloud gateway in the industry, of which a single cluster carries dozens of Tbps traffic, withstanding peak-hour traffic in large online shopping festivals.

1 INTRODUCTION

The public clouds provide customers with the ability to use on-demand, scalable computing resources (e.g., servers, storage, networking) to achieve agility, cost efficiency and business continuity based on cloud resource pooling and autoscaling [36]. Worldwide end-user spending on public cloud services has grown persistently over years [12] and the global pandemic has reinforced this trend [13]. To meet the increasing cloud usage demands, public cloud vendors, such as Amazon, Microsoft, Google and Alibaba, have invested lots of cutting-edge research and development efforts into addressing the performance, scalability, elasticity, stability and reliability of the globally distributed cloud infrastructures.

A cloud gateway is a packet forwarding device, which provides performant and reliable connectivity to globally distributed cloud resources in a multi-tenant environment [8, 25]. To satisfy the resource isolation requirements from customers, public cloud vendors build virtual private clouds (VPCs) [41] within the public cloud through overlay network protocols such as VXLAN [27]. Virtual machines (VMs) within the same VPC can communicate with each other while VMs belonging to different VPCs are isolated and transparent from each other. Sometimes, a VM in one VPC needs to com-

Jupiter Evolving: Transforming Google's Datacenter Network via Optical Circuit Switches and Software-Defined Networking

Google

Leon Poutievski, Omid Mashayekhi, Joon Ong, Arjun Singh, Mukarram Tariq,
Rui Wang, Jianan Zhang, Virginia Beauregard, Patrick Conner, Steve Gribble,
Rishi Kapoor, Stephen Kratzer, Nanfang Li, Hong Liu, Karthik Nagaraj,
Jason Ornstein, Samir Sawhney, Ryohei Urata, Lorenzo Vicisano, Kevin Yasumura,
Shidong Zhang, Junlan Zhou, Amin Vahdat
Google
sigcomm-jupiter-evolving@google.com

ABSTRACT

We present a decade of evolution and production experience with Jupiter datacenter network fabrics. In this period Jupiter has delivered 5x higher speed and capacity, 30% reduction in capex, 41% reduction in power, incremental deployment and technology refresh all while serving live production traffic. A key enabler for these improvements is *evolving Jupiter from a Clos to a direct-connect topology among the machine aggregation blocks*. Critical architectural changes for this include: A datacenter interconnection layer employing Micro-Electro-Mechanical Systems (MEMS) based Optical Circuit Switches (OCSes) to enable dynamic topology reconfiguration, centralized Software-Defined Networking (SDN) control for traffic engineering, and automated network operations for incremental capacity delivery and topology engineering. We show that the combination of traffic and topology engineering on direct-connect fabrics achieves similar throughput as Clos fabrics for our production traffic patterns. We also optimize for path lengths: 60% of the traffic takes direct path from source to destination aggregation blocks, while the remaining transits one additional block, achieving an average block-level path length of 1.4 in our fleet today. OCS also achieves 3x faster fabric reconfiguration compared to pre-evolution Clos fabrics that used a patch panel based interconnect.

KEYWORDS

Datacenter network, Software-defined networking, Traffic engineering, Topology engineering, Optical circuit switches.

ACM Reference Format:

Leon Poutievski, Omid Mashayekhi, Joon Ong, Arjun Singh, Mukarram Tariq, Rui Wang, Jianan Zhang, Virginia Beauregard, Patrick Conner, Steve Gribble, Rishi Kapoor, Stephen Kratzer, Nanfang Li, Hong Liu, Karthik Nagaraj, Jason Ornstein, Samir Sawhney, Ryohei Urata, Lorenzo Vicisano, Kevin Yasumura, Shidong Zhang, Junlan Zhou, Amin Vahdat Google sigcomm-jupiter-evolving@google.com . 2022. Jupiter Evolving: Transforming Google's Datacenter Network via Optical Circuit Switches and Software-Defined Networking. In *Proceedings of ACM Conference (SIGCOMM'22)*. ACM, New York, NY, USA, 20 pages. <https://doi.org/10.1145/3544216.3544265>

1 INTRODUCTION

Software-Defined Networking and Clos topologies [1, 2, 14, 24, 33] built with merchant silicon have enabled cost effective, reliable building-scale datacenter networks as the basis for Cloud infrastructure. A range of networked services, machine learning workloads, and storage infrastructure leverage uniform, high bandwidth connectivity among tens of thousands of servers to great effect.

While there is tremendous progress, managing the heterogeneity and incremental evolution of a building-scale

Scouts: Improving the Diagnosis Process Through Domain-customized Incident Routing

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ABSTRACT

Incident routing is critical for maintaining service level objectives in the cloud: the time-to-diagnosis can increase by 10× due to mis-routings. Properly routing incidents is challenging because of the complexity of today’s data center (DC) applications and their dependencies. For instance, an application running on a VM might rely on a functioning host-server, remote-storage service, and virtual and physical network components. It is hard for any one team, rule-based system, or even machine learning solution to fully learn the complexity and solve the incident routing problem. We propose a different approach using per-team Scouts. Each team’s Scout acts as its gate-keeper — it routes relevant incidents to the team and routes-away unrelated ones. We solve the problem through a collection of these Scouts. Our PhyNet Scout alone — currently deployed in production — reduces the time-to-mitigation of 65% of mis-routed incidents in our dataset.

service-level objectives. When incidents are *mis*-routed (sent to the wrong team), their time-to-diagnosis can increase by 10× [21].

A handful of well-known teams that underpin other services tend to bear the brunt of this effect. The physical networking team in our large cloud, for instance, is a recipient in 1 in every 10 mis-routed incidents (see §3). In comparison, the hundreds of other possible teams typically receive 1 in 100 to 1 in 1000. These findings are common across the industry (see Appendix A).

Incident routing remains challenging because modern DC applications are large, complex, and distributed systems that rely on many sub-systems and components. Applications’ connections to users, for example, might cross the DC network and multiple ISPs, traversing firewalls and load balancers along the way. Any of these components may be responsible for connectivity issues. The internal architectures and the relationships between these components may change over time. In the end, we find that the traditional

Project Silica: Towards Sustainable Cloud Archival Storage in Glass

Microsoft

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Abstract

Sustainable and cost-effective long-term storage remains an unsolved problem. The most widely used storage technologies today are magnetic (hard disk drives and tape). They use media that degrades over time and has a limited lifetime, which leads to inefficient, wasteful, and costly solutions for long-lived data. This paper presents Silica: the first cloud storage system for archival data underpinned by quartz glass, an extremely resilient media that allows data to be left in situ indefinitely. The hardware and software of Silica have been co-designed and co-optimized from the media up to the service level with sustainability as a primary objective. The design follows a cloud-first, data-driven methodology underpinned by principles derived from analyzing the archival workload of a large public cloud service. Silica can support a wide range of archival storage workloads and ushers in a new era of sustainable, cost-effective storage.

Keywords: Storage, Cloud Storage, Cold Storage, Disaggregation, Data Center, Archival, Sustainability, Glass

1 Introduction

Demand for archival storage for high-value, long-lived data continues to grow exponentially [26]. Cloud archival storage providers exploit statistical multiplexing of many customers' workloads onto a shared storage platform to offer the lowest-cost storage. Today, in all large-scale cloud providers, the most cost-effective media is magnetic, usually combining hard disk drives (HDDs) and tape. *All* magnetic media has limited lifetimes, and the data on it degrades over time, so it has to be migrated. In particular, HDDs have a ~5-year and tape a ~10-year lifetime. Since bit rot occurs in both over time, storage systems use scrubbing [5, 38, 39] to guarantee data integrity. Further, both have significant media costs, requiring garbage collection and defragmentation over time. As a result, while it

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How to Read a Paper

Version of February 17, 2016

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How to read a paper

- Golden rule: avoid as much as possible to drown in the details before getting a bird's-eye view
- Three-Pass approach
 - First pass aim: to get a general idea of the paper
 - Second pass aim: to grasp the paper's content but not its details
 - Third pass aim: to understand the paper in depth

The first pass

- A 5-10 minutes process to get a feeling of the paper
- You shall focus on:
 - Title
 - Abstract
 - Introduction
 - Headings
 - Conclusions

The first pass

- This pass will help you understand:
 - Category: is it a measurement paper? Is it an Operating System paper?
 - Context: which other papers is it related to?
 - Correctness: do the assumptions appear to be valid?
 - Contribution: what's the main contribution?
 - Clarity: is the paper well written?

The second pass

- A 1 hour process that allows you to grasp the content of the paper
- Read the paper with greater care but ignore details (e.g., proofs)
- This pass helps you to jot down the key points of the work
- After this pass you shall be able to summarize the main points of the paper, with supporting evidences, to someone else

The third pass

- This can take more than 1 hour and requires greater attention to details
- You should identify and challenge every assumption in every statement
- You should think about how you would present the same idea
 - This helps you figuring out how you would continue the work
- After this pass you shall be able to pinpoint even implicit assumptions and potential issues with experimental or analytical techniques

The End
(of the introductory lecture)

Questions?